MATH 202 Your name: _____

Learning target D3, version 1

D3: Apply Euler's Method to generate an approximate solution to an initial value problem, and explain the result's meaning in context.

A population of flamingos changes over time according to the differential equation

$$\frac{dP}{dt} = -\frac{1}{3}P(1-P)(5-P),$$

where P is measured in thousands (and t is probably measured in months or years).

(a) If $P(0) = 0.5$, use Euler's method with 3 steps to approximate $P(2)$. Record your work in a table. (Hint: If you're using 3 steps to get from $t = 0$ to $t = 2$, how big should Δt be?)

(b) Same thing, but this time, start with $P(0) = 1.5$.

(c) Compare your numerical results here with your graphical results on the D2 part of this problem. What recommendations do you have for flamingo keepers?

MATH 202 Your name: _____

Learning target D3, version 2

D3: Apply Euler's Method to generate an approximate solution to an initial value problem, and explain the result's meaning in context.

Consider the differential equation

$$\frac{dy}{dx} = -\frac{x}{y}$$

Starting with the initial condition $y(1) = 1$, use Euler's method with a step size of $\Delta x = 0.25$ to find out approximately when the solution $y(x)$ would cross over the x-axis.

Plot your points as you calculate them:

